

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Monday 9 February 2026

1.1 Daily Limit

Limit of a Rational Summation

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k}{n^2 + k^2}$$

Solution: We can evaluate this limit by recognizing it as a Riemann sum. First, factor out $\frac{1}{n^2}$ from the denominator of the summand:

$$S_n = \sum_{k=1}^n \frac{k}{n^2 \left(1 + \left(\frac{k}{n}\right)^2\right)} = \frac{1}{n} \sum_{k=1}^n \frac{\frac{k}{n}}{1 + \left(\frac{k}{n}\right)^2}$$

Taking the limit as $n \rightarrow \infty$, this precisely defines a definite integral over the interval $[0, 1]$. Letting $x_k = \frac{k}{n}$ with $\Delta x = \frac{1}{n}$, we obtain:

$$L = \int_0^1 \frac{x}{1 + x^2} dx$$

To evaluate this integral, let $u = 1 + x^2$, which implies $du = 2x dx$, or $x dx = \frac{1}{2} du$. The limits of integration change from $x \in [0, 1]$ to $u \in [1, 2]$:

$$L = \frac{1}{2} \int_1^2 \frac{1}{u} du = \frac{1}{2} [\ln |u|]_1^2$$

$$L = \frac{1}{2} (\ln 2 - \ln 1) = \frac{1}{2} \ln 2$$

Tuesday 10 February 2026

2.1 Daily Limit

Trigonometric Difference Limit

Evaluate the limit:

$$\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x} \right)$$

Solution: We combine the terms into a single rational expression:

$$L = \lim_{x \rightarrow 0} \frac{\sin^2 x - x^2}{x^2 \sin^2 x}$$

We will use Maclaurin series expansions to evaluate this indeterminate form. The expansion for $\sin x$ around $x = 0$ is:

$$\sin x = x - \frac{x^3}{6} + \mathcal{O}(x^5)$$

Squaring this expression yields:

$$\sin^2 x = \left(x - \frac{x^3}{6} \right)^2 = x^2 - \frac{x^4}{3} + \mathcal{O}(x^6)$$

Substitute this back into the numerator:

$$\sin^2 x - x^2 = \left(x^2 - \frac{x^4}{3} + \mathcal{O}(x^6) \right) - x^2 = -\frac{x^4}{3} + \mathcal{O}(x^6)$$

For the denominator, we use the dominant term of $\sin x$, which is x :

$$x^2 \sin^2 x \approx x^2(x^2) = x^4$$

Now, returning to the limit evaluation:

$$L = \lim_{x \rightarrow 0} \frac{-\frac{x^4}{3} + \mathcal{O}(x^6)}{x^4}$$

Factoring out x^4 and taking the limit gives:

$$L = -\frac{1}{3}$$

Wednesday 11 February 2026

3.1 Daily Limit

Trigonometric Exponential Limit

Evaluate the limit:

$$\lim_{x \rightarrow \infty} \left(\sin \frac{1}{x} + \cos \frac{1}{x} \right)^x$$

Solution: This limit is of the indeterminate form 1^∞ . Let us substitute $t = \frac{1}{x}$. As $x \rightarrow \infty$, $t \rightarrow 0^+$. The limit becomes:

$$L = \lim_{t \rightarrow 0^+} (\sin t + \cos t)^{\frac{1}{t}}$$

To handle the variable exponent, we apply the natural logarithm:

$$\ln L = \lim_{t \rightarrow 0^+} \frac{\ln(\sin t + \cos t)}{t}$$

This yields the indeterminate form $\frac{0}{0}$. We can apply L'Hôpital's Rule:

$$\ln L = \lim_{t \rightarrow 0^+} \frac{\frac{d}{dt} \ln(\sin t + \cos t)}{\frac{d}{dt}(t)}$$

$$\ln L = \lim_{t \rightarrow 0^+} \frac{\frac{\cos t - \sin t}{\sin t + \cos t}}{1}$$

Evaluating this limit directly by substituting $t = 0$:

$$\ln L = \frac{\cos(0) - \sin(0)}{\sin(0) + \cos(0)} = \frac{1 - 0}{0 + 1} = 1$$

Therefore, exponentiating both sides yields:

$$L = e^1 = e$$

Thursday 12 February 2026

4.1 Daily Limit

Sinc Function Exponential Limit

Evaluate the limit:

$$\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{1 - \cos x}}$$

Solution: As $x \rightarrow 0$, the base approaches 1 and the exponent approaches ∞ , producing the indeterminate form 1^∞ . Let the limit be denoted by L . Taking the natural logarithm gives:

$$\ln L = \lim_{x \rightarrow 0} \frac{\ln \left(\frac{\sin x}{x} \right)}{1 - \cos x}$$

We evaluate this using Maclaurin series expansions. For the sine function, $\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \mathcal{O}(x^4)$. Using the logarithmic approximation $\ln(1 + u) \approx u$ for small u , where $u = -\frac{x^2}{6}$:

$$\ln \left(\frac{\sin x}{x} \right) \approx \ln \left(1 - \frac{x^2}{6} \right) \approx -\frac{x^2}{6}$$

For the cosine function, the expansion is $\cos x = 1 - \frac{x^2}{2} + \mathcal{O}(x^4)$. Therefore:

$$1 - \cos x \approx 1 - \left(1 - \frac{x^2}{2} \right) = \frac{x^2}{2}$$

Substituting these approximations back into our logarithm limit expression:

$$\ln L = \lim_{x \rightarrow 0} \frac{-\frac{x^2}{6}}{\frac{x^2}{2}} = -\frac{1}{3}$$

Thus, exponentiating both sides gives the final limit:

$$L = e^{-1/3}$$

Friday 13 February 2026

5.1 Daily Limit

Indefinite Integral Limit

Evaluate the limit:

$$\lim_{x \rightarrow \infty} \int x^{\frac{1}{x}} \cdot \frac{1 - \ln(x)}{x^2} dx$$

Solution: First, we must evaluate the indefinite integral before taking the limit. We can find the antiderivative by recognizing the expression as the derivative of a particular function. Consider the function $f(x) = x^{\frac{1}{x}}$. We can rewrite this using the exponential function:

$$f(x) = e^{\frac{\ln x}{x}}$$

To find its derivative, we use the chain rule and the quotient rule:

$$\begin{aligned} f'(x) &= e^{\frac{\ln x}{x}} \cdot \frac{d}{dx} \left(\frac{\ln x}{x} \right) \\ \frac{d}{dx} \left(\frac{\ln x}{x} \right) &= \frac{\frac{1}{x} \cdot x - \ln x \cdot 1}{x^2} = \frac{1 - \ln x}{x^2} \end{aligned}$$

Therefore, the derivative is precisely the integrand:

$$f'(x) = x^{\frac{1}{x}} \cdot \frac{1 - \ln x}{x^2}$$

This means the evaluation of the integral gives us back the function:

$$\int x^{\frac{1}{x}} \cdot \frac{1 - \ln x}{x^2} dx = x^{\frac{1}{x}} + C$$

Ignoring the constant of integration for the purpose of the asymptotic limit analysis, we evaluate:

$$L = \lim_{x \rightarrow \infty} x^{\frac{1}{x}}$$

Let $y = \lim_{x \rightarrow \infty} x^{\frac{1}{x}}$. Taking the natural logarithm:

$$\ln y = \lim_{x \rightarrow \infty} \frac{\ln x}{x}$$

By L'Hôpital's Rule:

$$\ln y = \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0$$

Thus, $y = e^0 = 1$. Assuming the constant of integration vanishes or evaluating the fundamental functional behavior, the limit is:

$$L = 1$$

Saturday 14 February 2026

6.1 Daily Limit

Maclaurin Series Expansion Limit

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\ln(\cos x) + \frac{x^2}{2} + \frac{x^4}{12}}{x^6}$$

Solution: To evaluate this limit, we must expand $\ln(\cos x)$ up to the x^6 term. The Maclaurin series for $\cos x$ is:

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \mathcal{O}(x^8)$$

The Maclaurin series for $\ln(1 + u)$ is:

$$\ln(1 + u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \mathcal{O}(u^4)$$

Let $u = -\frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720}$. We substitute u into the logarithmic expansion: 1. $u = -\frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720}$
2. $u^2 = \left(-\frac{x^2}{2} + \frac{x^4}{24}\right)^2 \approx \frac{x^4}{4} - \frac{x^6}{24}$ 3. $u^3 = \left(-\frac{x^2}{2}\right)^3 \approx -\frac{x^6}{8}$

Now, combine these into the series for $\ln(1 + u)$:

$$\begin{aligned} \ln(\cos x) &= \left(-\frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720}\right) - \frac{1}{2} \left(\frac{x^4}{4} - \frac{x^6}{24}\right) + \frac{1}{3} \left(-\frac{x^6}{8}\right) \\ &= -\frac{x^2}{2} + x^4 \left(\frac{1}{24} - \frac{1}{8}\right) + x^6 \left(-\frac{1}{720} + \frac{1}{48} - \frac{1}{24}\right) \end{aligned}$$

Calculating the coefficients: For x^4 : $\frac{1}{24} - \frac{3}{24} = -\frac{2}{24} = -\frac{1}{12}$ For x^6 : $-\frac{1}{720} + \frac{15}{720} - \frac{30}{720} = -\frac{16}{720} = -\frac{1}{45}$ So, $\ln(\cos x) \approx -\frac{x^2}{2} - \frac{x^4}{12} - \frac{x^6}{45}$.

Substitute this expansion back into the numerator:

$$\text{Numerator} = \left(-\frac{x^2}{2} - \frac{x^4}{12} - \frac{x^6}{45}\right) + \frac{x^2}{2} + \frac{x^4}{12} = -\frac{x^6}{45}$$

Evaluating the limit:

$$L = \lim_{x \rightarrow 0} \frac{-\frac{x^6}{45}}{x^6} = -\frac{1}{45}$$

Sunday 15 February 2026

7.1 Daily Limit

Rational Function Parameters Limit

Let $f(x) = \frac{ax^2+bx+c}{bx^2+cx+d}$ where $a, b, c, d \in \mathbb{R}$ such that $f(0) = 0$, $f(\pi)$ DNE, and $\lim_{x \rightarrow \infty} f(x) = -\frac{1}{\pi}$. Find $\lim_{x \rightarrow \pi} f(x)$.

Solution: Let us systematically apply the given conditions to find the parameters of $f(x)$.

Condition 1: $f(0) = 0$ Substituting $x = 0$ yields $f(0) = \frac{c}{d} = 0$, meaning $c = 0$ (and $d \neq 0$). Our function simplifies to:

$$f(x) = \frac{ax^2 + bx}{bx^2 + d}$$

Condition 2: $\lim_{x \rightarrow \infty} f(x) = -\frac{1}{\pi}$ As $x \rightarrow \infty$, the limit of the rational function is the ratio of its leading coefficients:

$$\lim_{x \rightarrow \infty} f(x) = \frac{a}{b} = -\frac{1}{\pi} \implies b = -a\pi$$

Condition 3: $f(\pi)$ DNE For a rational function to not exist at $x = \pi$, the denominator must be zero at this point.

$$b(\pi)^2 + d = 0$$

Substituting $b = -a\pi$, we find:

$$(-a\pi)\pi^2 + d = 0 \implies -a\pi^3 + d = 0 \implies d = a\pi^3$$

Now substitute the expressions for b, c , and d back into the simplified $f(x)$:

$$f(x) = \frac{ax^2 + (-a\pi)x}{(-a\pi)x^2 + a\pi^3}$$

Factoring out a from the numerator and denominator (assuming $a \neq 0$):

$$f(x) = \frac{x^2 - \pi x}{-\pi x^2 + \pi^3} = \frac{x(x - \pi)}{-\pi(x^2 - \pi^2)}$$

We can factor the denominator as a difference of squares:

$$f(x) = \frac{x(x - \pi)}{-\pi(x - \pi)(x + \pi)}$$

For $x \neq \pi$, the common term $(x - \pi)$ cancels out:

$$f(x) = \frac{x}{-\pi(x + \pi)}$$

Evaluating the final limit: Now we find the limit as $x \rightarrow \pi$:

$$L = \lim_{x \rightarrow \pi} f(x) = \lim_{x \rightarrow \pi} \frac{x}{-\pi(x + \pi)}$$

$$L = \frac{\pi}{-\pi(\pi + \pi)} = \frac{\pi}{-\pi(2\pi)} = -\frac{1}{2\pi}$$

Monday 16 February 2026

8.1 Daily Limit

Trigonometric Radical Limit

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \frac{n}{2} \sqrt{2 - 2 \cos \left(\frac{2\pi}{n} \right)}$$

Solution: We utilize the half-angle trigonometric identity, which states that $1 - \cos(\theta) = 2 \sin^2 \left(\frac{\theta}{2} \right)$. By factoring out 2 inside the square root, we obtain:

$$2 - 2 \cos \left(\frac{2\pi}{n} \right) = 2 \left(1 - \cos \left(\frac{2\pi}{n} \right) \right)$$

Applying the half-angle identity with $\theta = \frac{2\pi}{n}$:

$$1 - \cos \left(\frac{2\pi}{n} \right) = 2 \sin^2 \left(\frac{\pi}{n} \right)$$

Thus, the expression under the square root becomes:

$$2 \left(2 \sin^2 \left(\frac{\pi}{n} \right) \right) = 4 \sin^2 \left(\frac{\pi}{n} \right)$$

Taking the square root (and noting that for large n , $\frac{\pi}{n}$ is in the first quadrant, so the sine is positive):

$$\sqrt{4 \sin^2 \left(\frac{\pi}{n} \right)} = 2 \sin \left(\frac{\pi}{n} \right)$$

Substituting this back into the original limit expression:

$$L = \lim_{n \rightarrow \infty} \frac{n}{2} \left(2 \sin \left(\frac{\pi}{n} \right) \right) = \lim_{n \rightarrow \infty} n \sin \left(\frac{\pi}{n} \right)$$

We can rewrite this by moving n to the denominator to utilize the fundamental limit $\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$:

$$L = \lim_{n \rightarrow \infty} \frac{\sin \left(\frac{\pi}{n} \right)}{\frac{1}{n}}$$

Multiply the numerator and denominator by π :

$$L = \pi \lim_{n \rightarrow \infty} \frac{\sin \left(\frac{\pi}{n} \right)}{\frac{\pi}{n}} = \pi(1) = \pi$$

Tuesday 17 February 2026

9.1 Daily Limit

Radical Trigonometric Difference Limit

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\sqrt{1 + \tan x} - \sqrt{1 + \sin x}}{x^3}$$

Solution: To eliminate the radicals in the numerator, we multiply the expression by its conjugate:

$$L = \lim_{x \rightarrow 0} \frac{(\sqrt{1 + \tan x} - \sqrt{1 + \sin x})(\sqrt{1 + \tan x} + \sqrt{1 + \sin x})}{x^3(\sqrt{1 + \tan x} + \sqrt{1 + \sin x})}$$

Simplifying the numerator as a difference of squares:

$$L = \lim_{x \rightarrow 0} \frac{(1 + \tan x) - (1 + \sin x)}{x^3(\sqrt{1 + \tan x} + \sqrt{1 + \sin x})} = \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3(\sqrt{1 + \tan x} + \sqrt{1 + \sin x})}$$

We can separate the limit into the product of two limits:

$$L = \left(\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} \right) \cdot \left(\lim_{x \rightarrow 0} \frac{1}{\sqrt{1 + \tan x} + \sqrt{1 + \sin x}} \right)$$

The second limit evaluates straightforwardly by direct substitution:

$$\lim_{x \rightarrow 0} \frac{1}{\sqrt{1 + 0} + \sqrt{1 + 0}} = \frac{1}{2}$$

For the first limit, we factor out $\sin x$ from the numerator:

$$\tan x - \sin x = \frac{\sin x}{\cos x} - \sin x = \sin x \left(\frac{1}{\cos x} - 1 \right) = \sin x \left(\frac{1 - \cos x}{\cos x} \right)$$

We substitute this back and group the terms into recognizable limit forms:

$$\lim_{x \rightarrow 0} \frac{\sin x(1 - \cos x)}{x^3 \cos x} = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \cdot \left(\frac{1 - \cos x}{x^2} \right) \cdot \left(\frac{1}{\cos x} \right)$$

Evaluating each fundamental component as $x \rightarrow 0$:

- $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$
- $\lim_{x \rightarrow 0} \frac{1}{\cos x} = 1$

Multiplying these values yields $(1) \cdot \left(\frac{1}{2}\right) \cdot (1) = \frac{1}{2}$ for the first limit. Combining both limits, the final evaluation is:

$$L = \left(\frac{1}{2}\right) \cdot \left(\frac{1}{2}\right) = \frac{1}{4}$$

Wednesday 18 February 2026

10.1 Daily Limit

Mixed Asymptotic Expansion Limit

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\cos x - \sqrt{1 - x^2}}{\sin^2(2x) + x^2}$$

Solution: We will utilize Maclaurin series expansions to evaluate both the numerator and the denominator near $x = 0$.

Numerator: The standard expansion for $\cos x$ is:

$$\cos x \approx 1 - \frac{x^2}{2} + \frac{x^4}{24}$$

For the radical term, we use the generalized binomial theorem $(1 + u)^{1/2} \approx 1 + \frac{1}{2}u - \frac{1}{8}u^2$, substituting $u = -x^2$:

$$\sqrt{1 - x^2} = (1 - x^2)^{1/2} \approx 1 - \frac{x^2}{2} - \frac{x^4}{8}$$

Subtracting the two expansions gives:

$$\begin{aligned} \cos x - \sqrt{1 - x^2} &\approx \left(1 - \frac{x^2}{2} + \frac{x^4}{24}\right) - \left(1 - \frac{x^2}{2} - \frac{x^4}{8}\right) \\ &= \frac{x^4}{24} + \frac{x^4}{8} = \frac{x^4}{24} + \frac{3x^4}{24} = \frac{4x^4}{24} = \frac{x^4}{6} \end{aligned}$$

Denominator: We substitute the dominant term of the sine expansion, $\sin u \approx u$:

$$\sin^2(2x) \approx (2x)^2 = 4x^2$$

So the entire denominator behaves as:

$$\sin^2(2x) + x^2 \approx 4x^2 + x^2 = 5x^2$$

Evaluating the Ratio: We substitute the leading asymptotic behaviors back into our limit:

$$L = \lim_{x \rightarrow 0} \frac{\frac{x^4}{6}}{5x^2} = \lim_{x \rightarrow 0} \frac{x^2}{30}$$

By direct substitution of $x = 0$, the limit evaluates strictly to zero:

$$L = 0$$